

# Scaled Bridge Structure Report : Bending Vibrations

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**Abstract**—The objective of the laboratory experiment was to investigate the vibration modes of a structure modeled as a **simply-supported beam**. Using a set of accelerometers, data on the vibration modes were collected and initially represented in the time domain. After performing various manipulations, Power Spectral Densities (PSDs) were calculated to identify the beam's natural frequencies. In the final step, the average damping ratios and frequencies for the stable modes were computed. Additionally, the mode shapes were determined by considering the phase and amplitude at each measured point.

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## 1. Experimental Setup

The structure is modeled as a simply-supported beam, with its geometries depicted in Figure 1. The positions of the accelerometers are arranged to avoid the node points of the structure. This arrangement ensures accurate detection of the vibration modes. Additionally, variations along the y-axis enable the detection of torque modes, which might otherwise be mistaken for bending modes. The geometrical characteristics of the beam under consideration are detailed in Table 1.

In Table 2, we outline the material properties of the beam currently under examination.

## 2. Data

|        |        |
|--------|--------|
| Length | 186 cm |
| Width  | 15 cm  |
| Height | 1.5 cm |

Table 1. Geometrical characteristics of the beam

|                     |                       |
|---------------------|-----------------------|
| Material            | Aluminium             |
| E (Young's Modulus) | 70 GPa                |
| $\rho$ (Density)    | $2700 \text{ kg/m}^3$ |

Table 2. Material properties of the beam

The sensors utilized in the experiment, along with their locations and the corresponding channels to which they were connected, are summarized in Table 3.

| Serial Number | Position x | Position y | Channel |
|---------------|------------|------------|---------|
| 5N 10543      | 23.5 cm    | +7.5 cm    | 1       |
| 5N 10546      | 41 cm      | -7.5 cm    | 2       |
| 5N 10544      | 75 cm      | 0 cm       | 3       |
| 5N 10545      | 110 cm     | +7.5 cm    | 4       |
| LW 75233      | 162.5 cm   | -7.5 cm    | 5       |
| 5N 38752      | 144 cm     | 0 cm       | 6       |
| 5N 38751      | 116 cm     | -7.5 cm    | 7       |
| 5N 38750      | 52 cm      | +7.5 cm    | 8       |

Table 3. Accelerometers used in the experimental test

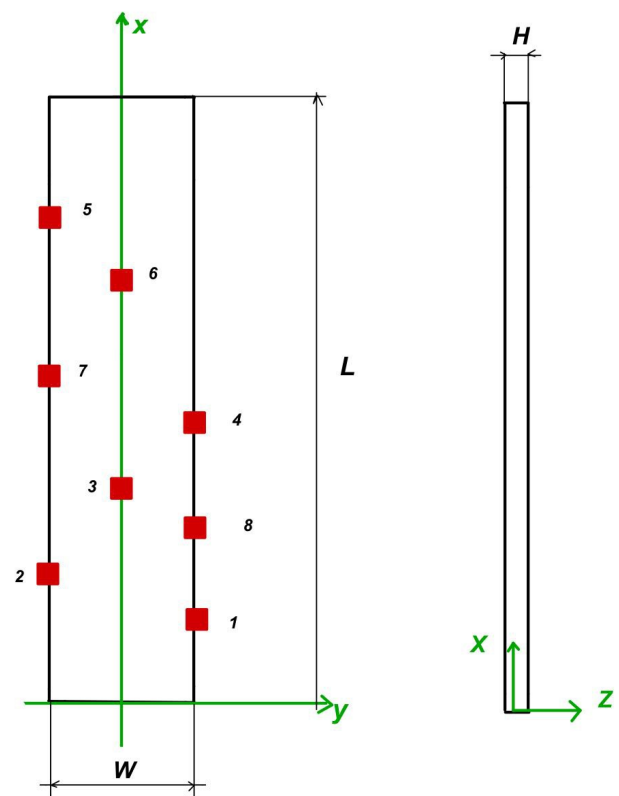


Figure 1. Experimental Setup , position of the accelerometers and schematic representation of the beam

## 3. Time Domain Measurements

Table 4 presents the three distinct types of measurements conducted throughout the laboratory session. Subsequent subsections will present the collected data in chronological order.

| Type | Sampling frequency [Hz] | Duration [s]    |
|------|-------------------------|-----------------|
| G3_1 | 256                     | 20              |
| G3_2 | 256                     | 20 <sup>1</sup> |
| G3_3 | 512                     | 20 <sup>2</sup> |

Table 4. Measurements in the Time Domain

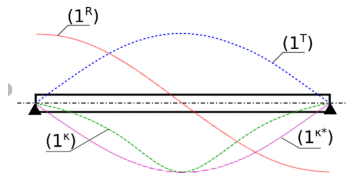
The data has been depicted in the time domain for every channel and measurement.

#### 4. Theoretical values

By using the **Euler-bernoulli theory**, we are able to compute the theoretical natural frequencies, with the data of our test through the formula 1.

In Fig. 2 we can see where are located the nodes, that should be avoided in the choice of the position for our accelerometers.

$$\omega_r = \frac{r^2 \pi^2}{L^2} \cdot \sqrt{\frac{E \cdot I}{\mu}} \quad (1)$$



**Figure 2.** Qualitative representation of the eigenfunctions, with the corresponding nodes for the first three modes

| MODE | Angular Frequency (rad/s) | Frequency (Hz) |
|------|---------------------------|----------------|
| 1    | 62.90                     | 10.01          |
| 2    | 251.59                    | 40.04          |
| 3    | 566.09                    | 90.10          |
| 4    | 1006.4                    | 160.17         |

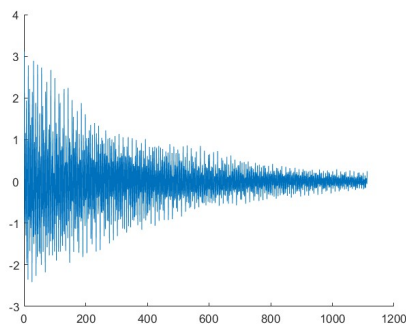
**Table 5.** Theoretical values of natural Frequencies and Corresponding Frequencies

#### 5. Measurements test

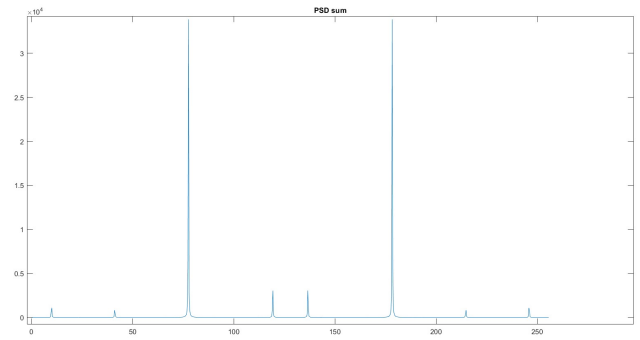
##### 5.1. First measurement

The sampling frequency chosen is  $f_s = 256\text{Hz}$ , Figure 3 shows the **impulse response function** that has been recorded, for in the first channel with in x axis the time. It is shown only the decade, after pre-processing process between two values:

- $n_1 = 1426$
- $n_2 = 2315$

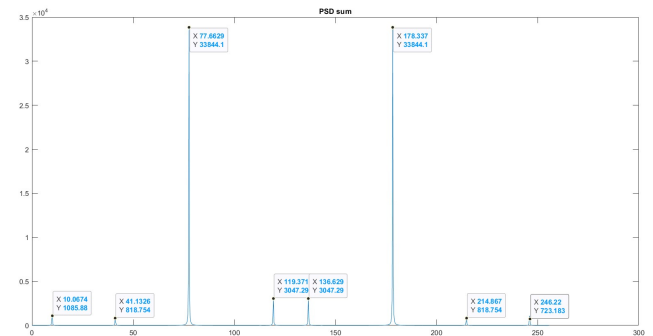


**Figure 3.** Impulse response function for the first emasurement in the first cable



**Figure 4.** Power spectral density for the first measurement

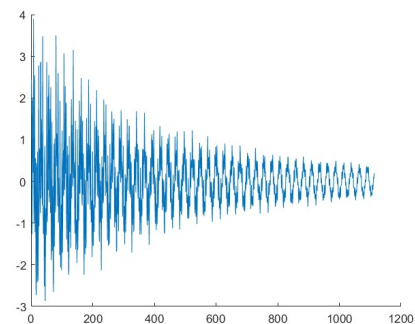
In figure 6 5 is shown the PSD as sum of the **Power Spectral Density** in each of the 8 channels, for the first measurement, we can observe the different peaks that corresponds in the x axis to the real natural frequencies measured.



**Figure 5.** Power spectral density for the first measurement with values of frequencies

Same considerations have been made for the second measurement and the third one as shown in the following;

##### 5.2. Second measurement



**Figure 6.** Impulse response function for the second measurement in the first cable

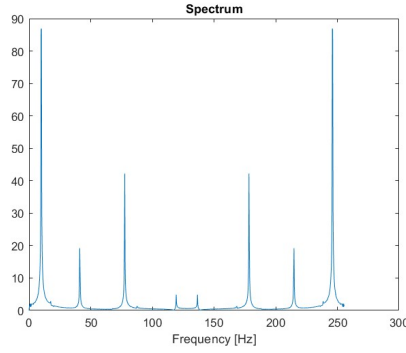


Figure 7. PSD for the second measurement

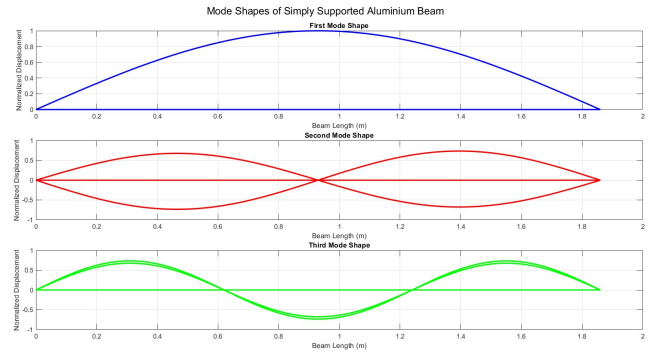


Figure 10. Stabilization diagram for the first measurement in the first cable with  $n = 8$

### 5.3. Third measurement

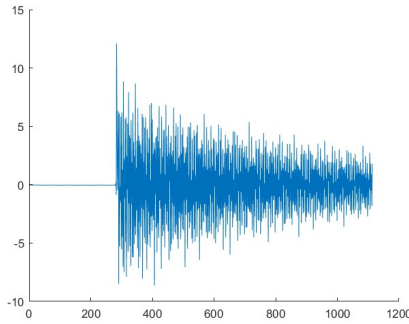


Figure 8. Impulse response function for the third measurement in the first cable

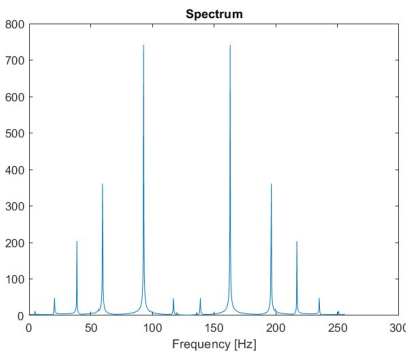


Figure 9. PSD for the third measurement

## 6. Mode Shapes

It is also possible to plot the graph of the first three mode shapes as shown in figure 10

## 7. PSD

To accurately identify the eigenfrequencies and derive precise estimates of the modal parameters, selecting the least noise-affected track is very important. Upon comparing the before-mentioned graphs, it becomes evident that the initial experiment offers the cleanest trend among the rest. **The Power Spectral Density (PSD)** is useful to measure the signal's energy level across various frequencies. Consequently, resonance peaks help us in preliminary identification of the eigenfrequencies' locations. Identifying these resonance points is crucial in determining the parameter 'n' for the complex exponential method. In this context, scrutinizing the PSD presented in Figure 8 reveals the presence of eight prominent peaks and so we can use this value as an attempt value for number of nodes, to check after with the stabilization diagram and CEM:  $n = 8$  Additionally, it's feasible to compute the theoretical frequencies utilizing an Euler-Bernoulli model of the beam, yielding and their values are summarized in Table 5.

$$\omega_r = \frac{r}{2\pi} \cdot \frac{1}{2} \sqrt{\frac{EI}{L^2}}$$

| MODE | REAL [Hz] |
|------|-----------|
| 1    | 10.07     |
| 2    | 41.13     |
| 3    | 77.66     |
| 4    | 119.37    |
| 5    | 136.63    |
| 6    | 178.34    |
| 7    | 214.87    |
| 8    | 245.93    |

Table 6. Frequencies evaluated in the PSD

## 8. Complex and

## Exponential Stabilization

## Method Diagram

The objective of the system identification phase is to apply the complex exponential method in order to estimate the modal parameters:

- $\omega_r$  eigenfrequency
- $\zeta_r$  damping coefficient
- $s_r$  pole

However, the method's reliability is not always assured, particularly when it yields solutions that are not practically viable. This occurs when the poles are not complex conjugates, or when the damping value is negative or too low. In this regard, the stabilization diagram

plays a crucial role as it helps distinguish between **computational solutions** lacking physical significance and those that are **stable** and acceptable.

It's essential to note that computational modes are independent of the system's physical characteristics. Instead, they result from a combination of numerical errors, measurement noise, and incorrect model ordering. Consequently, computational modes may vary randomly from the actual model order.

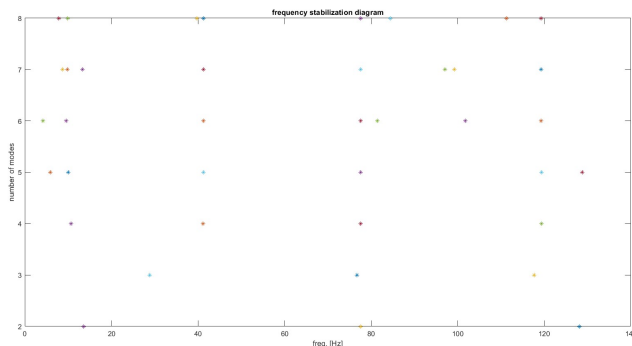
The stable modes manifest as vertical alignments at resonance peaks, underscoring the importance of accurately identifying these peaks. Subsequently, the following procedures were executed for each of the eight channels:

- Selection of  $n = 8$
- Implementation in MATLAB of the method
- Computation of the solutions

Using the stabilization diagram, stable solutions were identified for each of the eight vibration modes. An illustration of a stabilization diagram for the first channel is provided in Figure 11. This process can be replicated for all channels, leading to similar outcomes. The calculated modal parameters for each channel are summarized in Table 7, while Table 8 presents the average values of  $\omega_r$  and  $\zeta_r$  for each mode.

From the stabilization diagram and from the results of the COM method is possible to say that the number effective of nodes is 4 with corresponding four frequencies analysed.

In Table 7, notable are the instances of acceptable solutions highlighted in green, so the stable ones. The values of frequencies associated to damping ratio that are different from the others in stable conditions have been also neglected. A comparison with Table 6 reveals similarities in the first four eigenfrequencies values, reflecting the incorporation of resonance peak frequencies.



**Figure 11.** Stabilization diagram for the first measurement in the first cable with  $n = 8$

|   |                 |       |       |       |       |       |        |        |        |
|---|-----------------|-------|-------|-------|-------|-------|--------|--------|--------|
| 1 | $\omega_r$ [Hz] | 7.90  | 39.77 | 41.21 | 84.44 | 77.60 | 111.24 | 119.28 |        |
|   | $\zeta$ %       | 74.28 | 25.81 | 0.29  | 3.04  | 0.11  | 2.39   | 0.10   |        |
| 2 | $\omega_r$ [Hz] | 9.82  | 19.18 | 41.18 | 62.23 | 76.19 | 77.60  | 112.96 | 119.30 |
|   | $\zeta$ %       | 0.37  | 35.53 | 0.22  | 22.02 | 19.11 | 0.11   | 0.98   | 0.06   |
| 3 | $\omega_r$ [Hz] | 9.99  | 12.26 | 41.20 | 51.29 | 77.60 | 80.87  | 111.91 | 119.29 |
|   | $\zeta$ %       | 1.39  | 11.90 | 0.23  | 15.06 | 0.11  | 9.21   | 0.63   | 0.11   |
| 4 | $\omega_r$ [Hz] | 9.93  | 41.25 | 63.13 | 77.60 | 79.88 | 112.79 | 119.28 |        |
|   | $\zeta$ %       | 0.37  | 0.32  | 10.17 | 0.11  | 10.71 | 0.25   | 0.10   |        |
| 5 | $\omega_r$ [Hz] | 10.02 | 16.81 | 41.23 | 57.22 | 77.61 | 85.90  | 112.82 | 119.22 |
|   | $\zeta$ %       | 0.30  | 17.92 | 0.26  | 10.28 | 0.11  | 3.08   | 0.23   | 0.20   |
| 6 | $\omega_r$ [Hz] | 9.89  | 17.83 | 41.24 | 50.58 | 77.60 | 87.82  | 111.95 | 119.29 |
|   | $\zeta$ %       | 0.45  | 4.14  | 0.26  | 19.31 | 0.12  | 2.87   | 1.10   | 0.11   |
| 7 | $\omega_r$ [Hz] | 15.23 | 41.24 | 59.81 | 72.39 | 77.61 | 112.80 | 119.28 |        |
|   | $\zeta$ %       | 34.52 | 0.19  | 23.58 | 16.53 | 0.11  | 0.18   | 0.14   |        |
| 8 | $\omega_r$ [Hz] | 7.43  | 9.91  | 41.22 | 52.11 | 77.61 | 82.17  | 112.64 | 119.26 |
|   | $\zeta$ %       | 42.68 | 0.56  | 0.26  | 4.95  | 0.11  | 7.07   | 0.17   | 0.09   |

**Table 7.** Table with the values used in the COM method

Finally, by averaging the stable values of the frequencies and the corresponding damping ratios it is possible to identify the final values, which as seen are very similar to the values detected in the PSD, suggesting the correctness of the method of extrapolation of the modal parameters.

|                                     | mode 1 | mode 2 | mode 3 | mode 4 |
|-------------------------------------|--------|--------|--------|--------|
| Frequencies spectrum                | 10.07  | 41.13  | 77.66  | 119.37 |
| Average frequencies $\omega_r$ [Hz] | 9.91   | 41.22  | 77.60  | 119.27 |
| Average damping ratio $\zeta$ %     | 37%    | 25%    | 11%    | 11%    |

**Table 8.** Comparison Table of final results